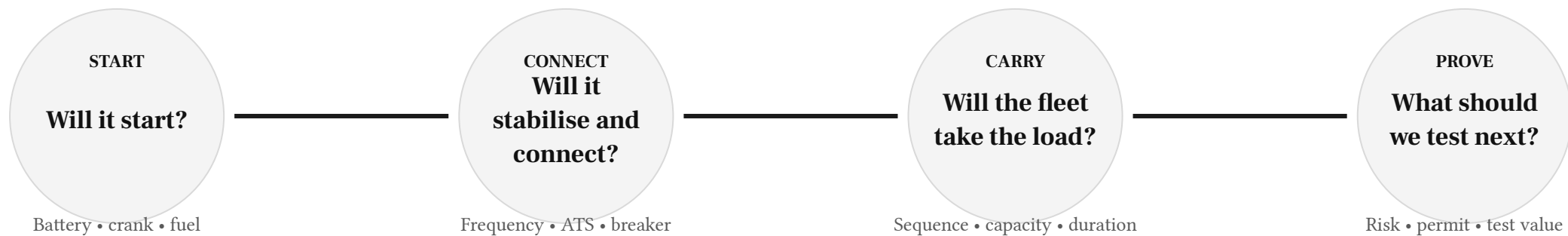


Know which part of emergency power may fail—and what to prove next

We tell the operations team which generator or shared emergency-power component is most likely to let them down, why, and what they should test or fix before the real outage.



Will it start?

Will it take the load?

Will the fleet work in sequence?

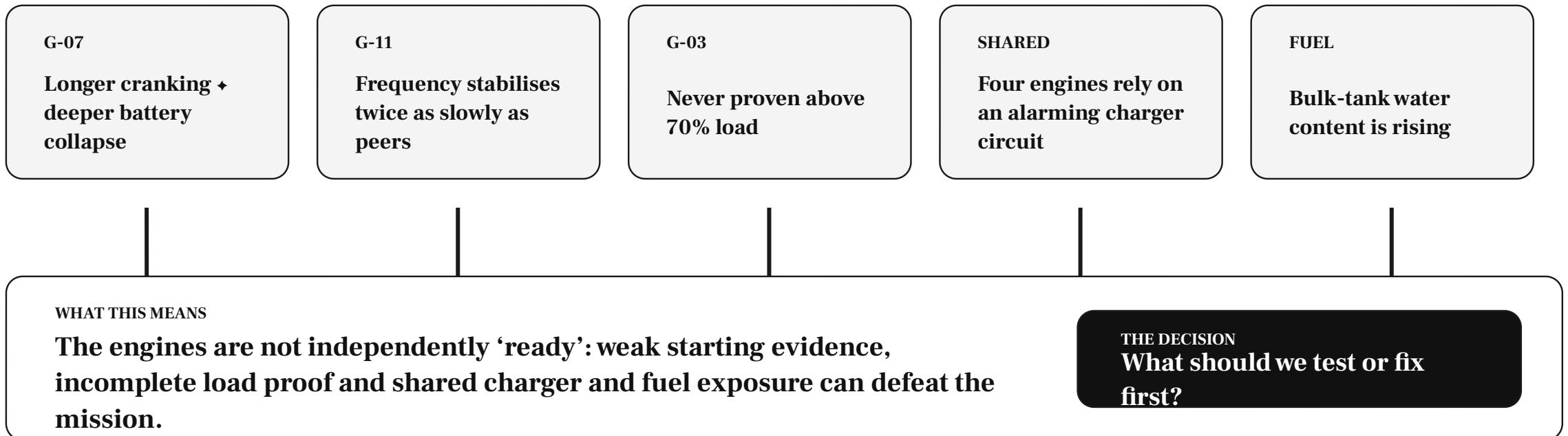
What must we prove next?

For critical-facilities, site-engineering, reliability and compliance teams deciding which asset needs testing or maintenance—and by when.

The fleet reports 100% start success—but readiness is conditional

SIXTEEN-GENERATOR SITE

Every generator passed its monthly start. The evidence still shows four different weaknesses.



The decision must consider failure risk, actual load, mandatory tests, permit hours and the approaching severe-weather window.

A cause-linked readiness and action queue—not another telemetry dashboard

CONDITIONALLY READY

G-07 needs targeted proof before the high-risk period.

Weak signal	Crank duration rose from 4.8 to 7.2 seconds
Supporting evidence	Minimum battery voltage is outside its peer range
Preparation	Inspect connections and charger output first
Recommended test	Controlled loaded start with high-resolution battery capture
Decision date	Complete before 5 September; retain mandatory intervals

SYSTEM READINESS

- Start** Battery, crank and fuel
- Stabilise** Speed, voltage and frequency
- Connect** ATS, breaker and protection
- Accept load** Step response and capacity
- Carry** Duration, cooling and fuel
- Recover** Transfer back and shutdown

The product recommends and explains. Facilities engineers authorise every test, maintenance task and operational action.

Spend scarce test, maintenance and permit hours where they reduce the most uncertainty

TODAY

A successful monthly start can create confidence without proving the mission.

- Engine start is treated as readiness
- Each generator is scored alone
- Pass/fail sheets hide deterioration
- Testing follows a generic cycle
- Permit use and resilience are separate

WITH RISK-BASED TEST PLANNING

Testing is targeted to the weakest evidence and the most important failure mode.

- ✓ Assess the complete start-to-load-and-recovery chain
- ✓ Expose shared fuel, charger, controls and switchgear risk
- ✓ Trend crank, battery, stability, load and peer behaviour
- ✓ Choose load, duration and instrumentation for a named gap
- ✓ Plan supplemental proof within mandatory and operating limits

Practical value: earlier weakness detection, focused maintenance, credible load proof, common-mode visibility and defensible readiness.

Reconstruct the full emergency-power mission—and the shared systems behind it

THE DATA WE NEED

1	Start evidence	Controller logs • high-resolution traces
2	Transfer sequence	ATS • switchgear • protection • PLC
3	Load proof	Exercise/load-bank records • real demand
4	Starting ♦ fuel	Battery • charger • tank • lab history
5	Asset history	Maintenance • alarms • interventions
6	Decision limits	Mandatory tests • permit and run hours

HOW THE ANSWER IS BUILT

- 1 Rebuild every event**
Align command, crank, stabilisation, breaker, load and shutdown
- 2 Test mission proof**
Apply approved start, transfer, load, duration and recovery criteria
- 3 Find the weak link**
Trend engines and components; map shared failure domains
- 4 Choose the most valuable action**
Rank inspect, maintain, instrument or test within constraints

Begin with evidence-backed rules and peer trends. Add calibrated probabilities only when sufficient witnessed failures and near misses exist.

What must be true before we test the use case

Start with one site, one complete start-to-load sequence and a bounded fleet history—not a broad predictive-maintenance build.

✓	Complete event history	Successful, failed, aborted, test and real-outage starts
✓	Transfer ♦ load evidence	ATS/switchgear sequence, actual load and duration
✓	Shared-system map	Fuel, chargers, controls, switchgear and ventilation
✓	Operating constraints	Mandatory tests, permit headroom and approved test window
✓	Qualified reviewers	Facilities engineers can validate failure mechanisms

If only monthly pass/fail sheets exist, position this as evidence improvement—not predictive readiness.

Plain language on the left; technical capability on the right

WHAT THE CLIENT HEARS	WHAT WE BUILD	WHAT IT DOES
● Read mixed-vendor events consistently	Multi-OEM normalisation	Map vendor states and alarms into one event language
● Replay the complete power sequence	Temporal reconstruction	Align utility, crank, frequency, breaker, load and recovery
● Check whether the mission was proven	Deterministic rules	Test approved start, transfer, load, duration and evidence criteria
● Find deterioration before failure	Signal + peer analysis	Detect crank, battery, stability, load and fuel weak signals
● See what can fail several engines	Dependency graph	Map shared fuel, charger, control and switchgear exposure
● Choose the right proof next	Constrained test model	Rank test value within safety, permit and mandatory rules

Control boundary: no autonomous start, stop, loading, transfer, setpoint change, OT write-back or removal of mandatory testing.